

# Lecture 14: The Factor Zoo

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# Intro

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- Last lecture: factor models worked, but interpretation was messy
- Today: maybe empirical success is the easy part
- LNS attacks the tests
- Harvey-Liu-Zhu attack the  $t$ -stats
- Kelly-Pruitt-Su change the object

- Last lecture: factor models worked, but interpretation was messy
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- Main theme: the data are high-dimensional, our tests are not

- Part I: why Lewellen, Nagel, and Shanken ([2010](#)) changed the standards for factor tests
- Part II: how Lewellen, Nagel, and Shanken ([2010](#)) tells us to raise the bar
- Part III: how Fama and French ([2015](#)) expands the benchmark
- Part IV: why Harvey, Liu, and Zhu ([2016](#)) turns the factor zoo into an inference problem
- Part V: how Kelly, Pruitt, and Su ([2019](#)) moves from factors to structure

## **Part I: We Should Have Higher Standards...**

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# The Factor-Zoo Explosion

During the 2000s and early 2010s:

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- Examples: accruals, profitability, investment, asset growth, ...
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- Also: several papers proposing theories + factors that price the 25 portfolios
- Example: Lettau and Ludvigson ([2001](#))



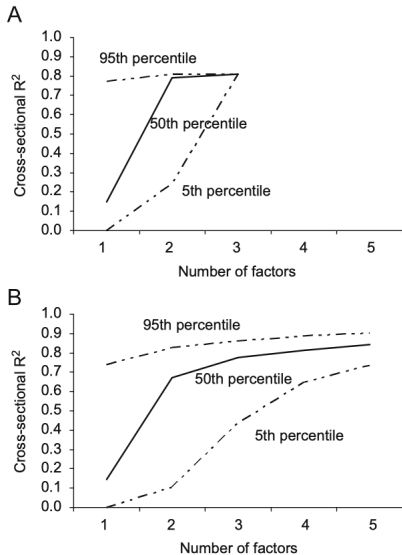
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- Example: Lettau and Ludvigson ([2001](#))
- Lewellen, Nagel, and Shanken ([2010](#)): pricing the 25 portfolios is too easy!

- The 25 portfolios have a very strong factor structure
- We already know that MKT, SMB, and HML price them well
- If you give me a factor that is correlated with them... it will price the 25 portfolios too!
- We're judging models based on just a few assets!

# A Killer Simulation



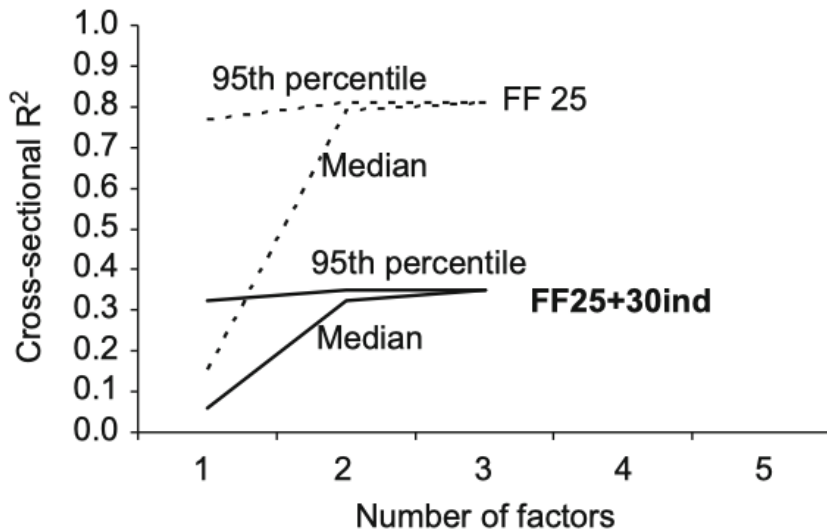
- Panel A: simulate random combinations of MKT, SMB, and HML
- Estimate the time-series betas and then run the cross-sectional pass
- Repeat many times and keep track of  $R^2$
- Panel B: do the same, but with combinations of the 25 portfolios
- If you're not scared, look again

## How To Improve Tests

They provide a few ways to improve the tests:

1. Expand the set of test portfolios beyond size-B/M portfolios
2. Take the magnitude of the cross-sectional slopes seriously
3. Report the GLS  $R^2$  instead of OLS  $R^2$
4. If a factor is traded, it should price itself (i.e., have a significant risk premium)
5. Report confidence intervals for the cross-sectional  $R^2$

## One Example of Improvement



Questions?

## Part III: Fama-French Reloaded

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# The Profitability and Investment Anomalies

- Titman, Wei, and Xie (2004): firms that invest more heavily have **lower** subsequent returns
- This is true controlling for size, B/M, and momentum
- Novy-Marx (2013): firms with higher profitability have **higher** subsequent returns
- The 3-factor model cannot explain these anomalies
- Fama and French (2006) had already noticed these two anomalies, with different implementations



## This Actually Has Some Interpretation

Recall the basic pricing formula:

$$P_t = \sum_{i=1}^{\infty} \mathbb{E}_t \left[ \frac{D_{t+i}}{(1+r)^{t+i}} \right]$$

- A firm with earnings  $Y_{t+j}$  that pays dividends  $D_{t+j}$  has book equity  $B_{t+j} = B_{t+j-1} + Y_{t+j} - D_{t+j}$
- The valuation formula becomes:

$$P_t = \sum_{i=1}^{\infty} \mathbb{E}_t \left[ \frac{Y_{t+i} - (B_{t+i} - B_{t+i-1})}{(1+r)^{t+i}} \right]$$

- Retained earnings (changes in book equity) are typically used for investment
- Equipment, R&D, marketing, etc...

## FF (2015): Extending the Benchmark

- These two anomalies were well-known and hard to explain with the 3-factor model
- Fama and French (2015): add two more factors to capture profitability and investment
- This is a direct extension of Fama and French (1993)... and a certain “concession”

## FF (2015): Extending the Benchmark

- These two anomalies were well-known and hard to explain with the 3-factor model
- Fama and French (2015): add two more factors to capture profitability and investment
- This is a direct extension of Fama and French (1993)... and a certain “concession”
- Two new factors created similarly to SMB and HML, but sorting on profitability and investment
- Profitability: *robust minus weak* (RMW)
- Investment: *conservative minus aggressive* (CMA)

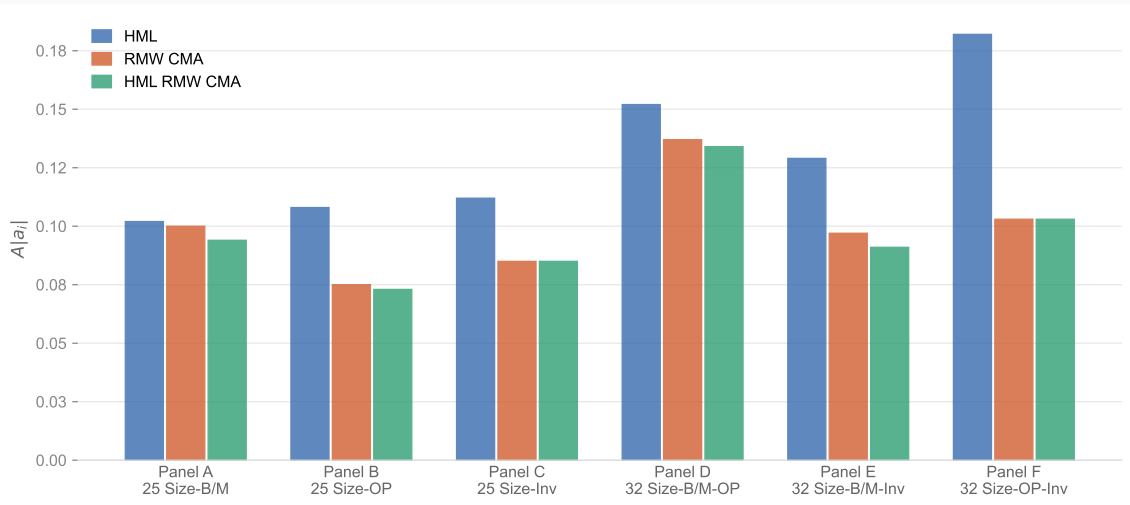
# The Five Factors

- MKT, SMB, and HML are measured as in Fama and French ([1993](#))
- Operating profitability: think about EBITDA
- Investment: asset growth

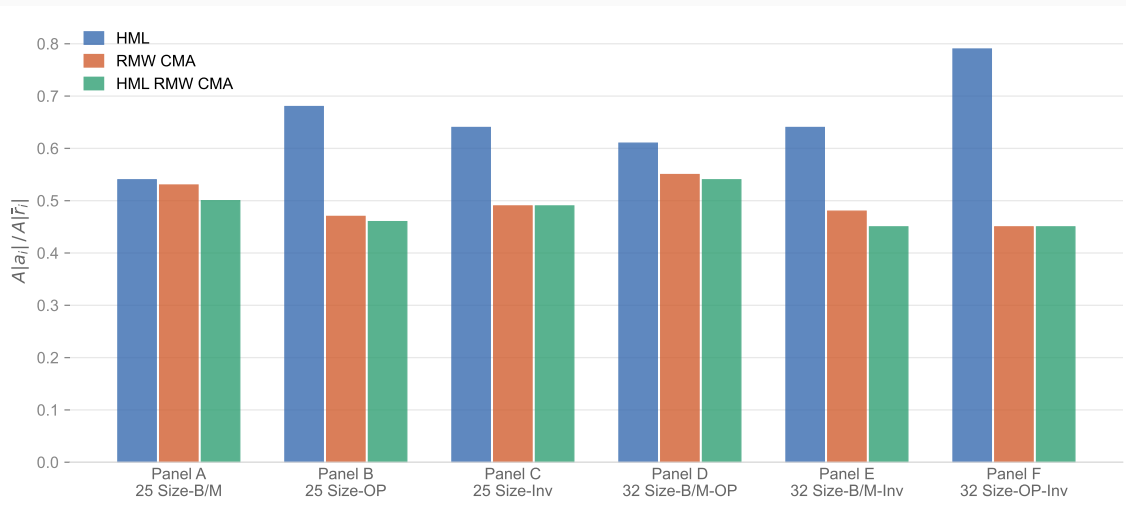
# The Five Factors

- MKT, SMB, and HML are measured as in Fama and French (1993)
- Operating profitability: think about EBITDA
- Investment: asset growth
- Create 2x2 sorts of (size, other variable), where other variable = B/M, profitability, or investment
- They study many ways to construct RMW and CMA and results won't change
- RMW and CMA are just zero-cost long-short portfolios, like SMB and HML

## Mean Absolute Intercepts – (MKT + SMB + ...)



# Mean Absolute Intercepts/Mean Absolute Returns – (MKT + SMB + ...)



## Fama-French (2015): Two Main Empirical Lessons

- Adding these factors did decrease the average  $\alpha$
- HML becomes somewhat redundant
- Vast majority of cases: the model fails the GRS test from Gibbons, Ross, and Shanken (1989)
- What does that mean?



## Fama-French (2015): Two Main Empirical Lessons

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- What does that mean?

Asset pricing models are simplified propositions about expected returns that are rejected in tests with power. We are less interested in whether competing models are rejected than in their relative performance, which we judge using GRS and other statistics. We want to identify the model that is the best (but imperfect) story for average returns on portfolios formed in different ways.



- Many anomalies explained
- Some survive
- Model performance depends on specification

*“Even benchmark models are unstable”*

## Part IV: The Factor Zoo

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- Hundreds of factors proposed
- Empirical success became cheap

- Multiple testing is the core problem
- Standard  $t$ -statistics are not enough

- List of factors (large table)



- Not reliable with many tests

- Distribution of t-stats / p-values

- Bonferroni
- Holm
- BHY

- Many factors lose significance

- Unknown number of attempted factors
- Publication bias

- LNS: weak tests
- FF: more factors
- Harvey: too many factors

*“What are we estimating?”*

## Part V: From Factors to Structure

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- Unify characteristics and factor models

- Characteristics summarize covariances
- Betas are functions of characteristics

# The Core Problem

- Low-dimensional models
- High-dimensional data
- High-dimensional beta representation

- Characteristics  $\rightarrow$  betas  $\rightarrow$  returns

# Factors Are Projections

- Factors are projections
- Not primitive objects

- Characteristics matter
- But via covariances

*“The factor zoo reflects high-dimensional structure”*

- Identification is central
- Adding factors is not a solution
- The problem is high-dimensional



# The Next Step

- Regularization
- Machine learning

Questions?

## Readings and References i

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